

LANCASTER FOX, CA, USA

WMO: 723816

Lat: 34.741N

Lon: 118.212W

Elev: 713

StdP: 93.05

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 03159

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -5.9	(c) -3.9	(d) -19.3	(e) 0.7	(f) 3.9	(g) -17.0	(h) 0.9	(i) 6.0	(j) 14.7	(k) 10.5	(l) 12.9	(m) 10.9	(n) 1.3	(o) 270	(p) 0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.1	(c) 39.7	(d) 18.7	(e) 38.3	(f) 18.1	(g) 37.0	(h) 17.6	(i) 20.3	(j) 35.5	(k) 19.6	(l) 34.9	(m) 18.8	(n) 34.3	(o) 6.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.6	26.7	13.3	10.4	27.2	12.0	9.5	26.4	61.2	34.9	58.4	35.0	55.7	34.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.5	12.1	11.2	-9.3	42.7	2.2	1.1	-10.9	43.5	-12.2	44.1	-13.5	44.7	-15.1	45.5		
			-11.1	21.7	2.0	1.0	-12.5	22.5	-13.7	23.1	-14.9	23.6	-16.4	24.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	7.5	8.9	12.2	15.0	19.8	24.9	28.5	27.7	23.5	17.1	10.9	6.7
	DBStd	8.35	3.18	3.27	3.58	4.05	4.47	4.20	2.73	2.52	3.37	3.65	3.73	3.34
	HDD10.0	313	89	54	19	6	0	0	0	0	0	1	33	112
	HDD18.3	1602	336	264	191	113	38	5	0	0	4	68	223	360
	CDD10.0	2847	12	24	88	157	302	448	572	548	406	220	61	9
	CDD18.3	1093	0	0	2	14	82	203	314	290	159	28	1	0
	CDH23.3	15595	1	8	98	315	1093	2683	4350	4062	2286	642	58	0
	CDH26.7	8786	0	1	17	98	505	1524	2648	2479	1267	242	5	0
Wind	WSAvg	4.9	3.3	4.2	5.1	6.3	6.7	6.6	5.8	5.1	4.4	4.1	3.6	3.5
Precipitation	PrecAvg	190	30	49	36	10	3	1	3	2	5	8	13	28
	PrecMax	367	116	158	141	46	28	21	24	26	32	61	53	136
	PrecMin	47	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	91	29	53	37	13	7	4	6	6	9	13	15	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	24.4	29.1	32.3	36.7	40.8	42.0	41.1	38.4	33.9	27.4	21.2
		MCWB	9.9	10.6	12.8	14.7	17.0	18.4	19.9	19.7	18.1	15.7	12.2	10.8
	2%	DB	19.0	22.1	26.1	29.5	34.1	38.0	40.0	39.4	36.9	31.6	25.1	18.5
		MCWB	8.6	10.0	12.0	13.3	15.8	17.4	19.0	18.6	17.5	14.6	11.2	8.9
	5%	DB	17.3	19.6	23.8	27.2	32.0	36.3	38.4	38.0	35.3	29.4	22.6	16.4
		MCWB	8.2	9.1	11.1	12.5	15.1	16.9	18.3	18.0	16.8	13.6	10.4	7.8
	10%	DB	15.1	16.9	21.3	24.5	29.5	34.4	36.9	36.6	33.7	27.2	19.9	14.3
		MCWB	7.3	8.2	10.3	11.7	14.2	16.3	17.9	17.4	16.1	12.7	9.3	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.5	14.0	15.8	18.0	19.6	21.6	21.3	20.3	17.7	13.6	12.5
		MCDB	17.6	19.2	25.4	28.7	33.8	37.4	36.5	36.8	34.5	27.0	22.7	17.5
	2%	WB	10.8	11.1	12.9	14.4	16.8	18.6	20.6	20.1	19.1	15.8	12.3	11.0
		MCDB	15.3	19.0	23.3	26.8	32.1	35.5	35.6	35.8	33.1	27.8	21.4	15.7
	5%	WB	9.6	10.1	11.9	13.3	15.7	17.8	19.9	19.2	18.1	14.8	11.3	9.7
		MCDB	14.4	17.4	21.7	24.6	30.1	34.1	34.8	35.5	31.9	26.5	20.1	13.9
	10%	WB	8.5	9.2	10.9	12.3	14.9	16.9	19.0	18.4	17.0	13.7	10.3	8.2
		MCDB	13.6	15.5	19.6	22.8	28.1	32.4	33.9	34.7	30.9	24.7	18.5	13.0
Mean Daily Temperature Range	5% DB	MDBR	14.8	14.1	15.0	14.8	14.7	15.7	16.1	17.4	18.1	17.9	16.7	14.7
		MCDBR	19.2	20.1	20.3	19.8	18.6	18.2	18.4	19.3	20.6	21.8	21.2	18.4
		MCWBR	11.1	10.6	9.5	8.6	7.4	6.8	6.8	7.6	8.5	10.1	11.1	10.9
	5% WB	MCDBR	13.5	15.9	17.0	16.9	17.2	17.1	15.8	17.0	17.8	18.2	17.2	13.2
		MCWBR	8.3	8.7	8.3	7.6	7.0	6.6	6.2	7.3	7.7	8.5	9.2	8.4
Clear-Sky Solar Irradiance	taub	0.292	0.302	0.320	0.347	0.361	0.355	0.382	0.362	0.347	0.332	0.299	0.292	
	taud	2.504	2.491	2.432	2.358	2.342	2.373	2.342	2.374	2.412	2.434	2.509	2.523	
	Ebn at Noon	923	949	955	937	922	923	896	911	912	900	905	901	
	Edh at Noon	82	94	108	122	126	122	125	119	109	98	81	77	
All-Sky Solar Radiation	RadAvg	3.08	3.98	5.43	6.71	7.58	8.28	7.89	7.42	6.29	4.73	3.52	2.81	
	RadStd	0.24	0.44	0.31	0.32	0.38	0.36	0.33	0.19	0.15	0.27	0.15	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) -1.92	(g) N/A	(h) N/A	(i) +1.18	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (46 neighbors)	+0.63	N/A	-1.78	+0.66	N/A	N/A	N/A	-115	+223	+106

Nomenclature: See separate page